HW 9

Monday, 20 March 2023 4:00 PM

Question 1

Steady state, non-isothermal catalyst pellet ; First order reaction

$$\nabla^2 u = \phi^2 u \exp\left\{\frac{\gamma\beta(1-u)}{1+\beta(1-u)}\right\} \qquad 0<\rho<1 \qquad ; \quad \nabla^2 = \frac{1}{\rho^2}\frac{d}{d\rho}\left(\rho^2\frac{d}{d\rho}\right)$$

(i) $u_N = 1 + (1-\rho^2)\sum_{n=0}^{N-1} \alpha_n^{(N)} P_n(\rho^2)$

$$u_0 = 1 + (1-\rho^2)\sum_{n=0}^{0} \alpha_n^{(N)} P_n(\rho^2) = 1 + (1-\rho^2)\left(\alpha_0 P_0(\rho^2)\right) \qquad ; \quad P_0(\rho^2) = (-1)^0 \frac{0!}{(0!)}\frac{\Gamma\left(\frac{1}{2}q+\frac{3}{2}+0\right)}{\Gamma\left(\frac{1}{2}q+\frac{3}{2}+0\right)}\frac{\Gamma\left(\frac{1}{2}q+\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}q+\frac{1}{2}+0\right)}\rho^{2(0)}$$

$$P_0(\rho^2) = 1$$

$$\therefore u_1 = 1 + (1-\rho^2)(\alpha_0) = 1 + \alpha_0 - \alpha_0\rho^2$$

$$P_1(\rho^2) = \sum_{k=0}^{1} (-1)^k \frac{(n!)}{(n-k)!}\frac{\Gamma\left(\frac{1}{2}q+\frac{3}{2}+n+k\right)}{\Gamma\left(\frac{1}{2}q+\frac{3}{2}+n\right)}\frac{\Gamma\left(\frac{1}{2}q+\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}q+\frac{1}{2}+k\right)}\rho^{2k}$$

$$= 1\cdot 1\cdot \frac{\Gamma\left(\frac{1}{2}q+\frac{3}{2}+1\right)\Gamma\left(\frac{1}{2}q+\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}q+\frac{3}{2}+1\right)\Gamma\left(\frac{1}{2}q+\frac{1}{2}+0\right)}\rho^0 + \frac{(-1)(1)\ \Gamma\left(\frac{1}{2}q+\frac{3}{2}+2\right)\Gamma\left(\frac{1}{2}q+\frac{1}{2}\right)}{\Gamma\left(\frac{1}{2}q+\frac{3}{2}+1\right)\Gamma\left(\frac{1}{2}q+\frac{1}{2}+1\right)}\rho^2$$

$$= 1 - \frac{\Gamma\left(\frac{q}{2}+\frac{7}{2}\right)\Gamma\left(\frac{q}{2}+\frac{1}{2}\right)}{\Gamma\left(\frac{q}{2}+\frac{5}{2}\right)\Gamma\left(\frac{q}{2}+\frac{3}{2}\right)}\rho^2$$

We know, $\Gamma(n) = (n-1)!$

$$\therefore \Gamma\left(\frac{q+7}{2}\right) = \left(\frac{q+7}{2}-1\right)! \quad \Rightarrow \quad = \left(\frac{q+5}{2}\right)! = \frac{(q+5)!}{2!}$$

$$\Gamma\left(\frac{q+1}{2}\right) = \frac{(q-1)!}{2!}$$

$$\Gamma\left(\frac{q+5}{2}\right) = \frac{(q+3)!}{2!}$$

$$\Gamma\left(\frac{q+3}{2}\right) = \frac{(q+1)!}{2!}$$

$$P_1(\rho^2) = 1 - \left[\frac{\frac{(q+5)!}{2}\cdot\frac{(q-1)!}{2}!}{\frac{(q+3)!}{2}\cdot\frac{(q+1)!}{2}!}\right]\rho^2$$

$$\frac{q+5}{2}-1 = \frac{q+3}{2}$$

$$\frac{q-1}{2}-1 = \frac{q-3}{2}$$

$$\frac{q+1}{2}-1 = q-1$$

$$P_1(\rho^2) = 1 - \frac{\frac{(q+5)}{2}\frac{(q+3)}{2}!\ \frac{(q-1)}{2}!}{\frac{(q+3)}{2}!\ \frac{(q+1)}{2}\frac{(q-1)}{2}!}\rho^2$$

$$P_1(\rho^2) = 1 - \frac{q+5}{(q+1)}\rho^2$$

From the question: collocation for N-point is the root of $(N+1)^{th}$ basis function

Root of: $P_1(\rho^2) = 1 - \frac{(q+5)}{(q+1)}\rho^2 = 0$

$$\rho^2 = \frac{(q+1)}{(q+5)}$$

$$\boxed{\rho^* = \left[\frac{(q+1)}{(q+5)}\right]^{1/2}}$$

(ii) $\eta = \frac{(q+1)}{\phi^2}\left.\frac{du}{d\rho}\right|_{\rho=1}$; $\frac{du_1}{d\rho} = -2\alpha_0\rho$

$$\left.\frac{du_1}{d\rho}\right|_{\rho=1} = -2\alpha_0$$

Solving for α_0: $u_1 = 1 + (1-\rho^2)\alpha_0$ @ $\rho = \rho^* \Rightarrow u_1 = U$

$$(U-1) = (1-\rho^{*2})\alpha_0$$

$$\alpha_0 = \frac{(U-1)}{(1-\rho^{*2})} = \frac{(U-1)}{\left[1-\frac{(q+1)}{(q+5)}\right]} = \frac{(U-1)(q+5)}{4}$$

$$\therefore \quad \eta = \frac{(q+1)}{\phi^2}(-2\alpha_0)$$

$$= \frac{(q+1)}{\phi^2}\frac{(-2)(U-1)(q+5)}{4}$$

$$\boxed{\eta = \frac{(q+1)(q+5)(1-U)}{2\phi^2}}$$

(iii) From the transport equation: $\nabla^2 u = \phi^2 u \exp\left\{\frac{\gamma\beta(1-u)}{1+\beta(1-u)}\right\}$

$$\nabla^2 u_1 = \nabla^2(1 + (1-\rho^2)\alpha_0)$$

$$= \frac{1}{\rho^q}\frac{d}{d\rho}\left(\rho^q \frac{d}{d\rho}[1 + (1-\rho^2)\alpha_0]\right)$$

$$= \frac{1}{\rho^q}\frac{d}{d\rho}\left(\rho^q \cdot (-2\rho)\alpha_0\right) = \frac{1}{\rho^q}\frac{d}{d\rho}\left(-2\rho^{(q+1)}\alpha_0\right)$$

$$= \frac{1}{\rho^q}(-2\alpha_0)(q+1)\rho^q$$

$$= -2\frac{(U-1)(q+5)(q+1)}{4} = \frac{(1-U)(q+5)(q+1)}{2}$$

@ $\rho = \rho^*$, $u_1 = u = U$

$$\therefore \nabla^2 u_1 = \phi^2 U \exp\left\{\frac{\gamma\beta(1-U)}{1+\beta(1-U)}\right\}$$

$$\boxed{\frac{(1-U)(q+5)(q+1)}{2} = \phi^2 U \exp\left\{\frac{\gamma\beta(1-U)}{1+\beta(1-U)}\right\}}$$

(iv) $q = 1$

$$(1-U)(6) = \phi^2 U \exp\left\{\frac{\gamma\beta(1-U)}{1+\beta(1-U)}\right\} \Rightarrow U \text{ as } f(\gamma, \beta, \phi^2)$$

η @ $q = 1$:

$$\eta = \frac{(q+1)(q+5)(1-U)}{2\phi^2} = \frac{6(1-U)}{\phi^2}$$

$$\eta\phi^2 = 6(1-U)$$

$$\frac{d(\eta\phi^2)}{d\phi} = \frac{d}{d\phi}(6-6U) = -6\frac{dU}{d\phi}$$

Using python, for a given γ, β value we can solve for η as a function of ϕ

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This can be plotted for various initial guesses for implicit function of U

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$\underline{\phi}$ & $\bar{\phi}$ represent the finite values for which multiplicity may appear

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Numerically it is shown in the graph

→ See python

Numerically it is shown in the graph

$\underline{\phi} = 0.3$; $\bar{\phi} = 0.6$ for $\gamma = 4$, $\beta = 10$ } See python file attached

Solving for the criteria: $\nabla^2 u = \phi^2 u \exp\left\{\frac{\gamma\beta(1-u)}{1+\beta(1-u)}\right\}$

If solution is not unique: $\underline{u} \le u \le \bar{u}$ everywhere in Ω

By letting $f(u) = u \exp\left\{\frac{\gamma\beta(1-u)}{1+\beta(1-u)}\right\}$

$\nabla^2 \bar{u} = -\phi^2 f(\bar{u})$ in Ω, $\bar{u} = 1$ on $\partial\Omega$

$\nabla^2 \underline{u} = -\phi^2 f(\underline{u})$ in Ω, $\underline{u} = 1$ on $\partial\Omega$

$$\int_\Omega \left[(\underline{u}-1)\nabla^2\bar{u} - (\bar{u}-1)\nabla^2\underline{u}\right] dV = -\phi^2 \int_\Omega \left[(\underline{u}-1) f(\bar{u}) - (\bar{u}-1) f(\underline{u})\right] dV$$

$$F(u) = \frac{f(u)}{(u-1)}$$

$$\Rightarrow -\phi^2 \int_\Omega (\underline{u}-1)(\bar{u}-1)F(\bar{u}) - (\bar{u}-1)(\underline{u}-1)F(\underline{u})\ dV$$

$$\Rightarrow \boxed{-\phi^2 \int_\Omega \left[(\underline{u}-1)(\bar{u}-1)\right]\left(F(\bar{u}) - F(\underline{u})\right) dV = 0}$$

Sufficient condition for uniqueness regardless of ϕ^2 value is $F(u)$ has to monotonic in range $0 \le u \le 1$; Additionally, this equation is a requirement for multiplicity

$\therefore \frac{dF}{du} < 0$, $0 \le u \le 1$

Differentiating, we get: $\frac{dF}{du} = \frac{d}{du}\left(\frac{f(u)}{(u-1)}\right) = \frac{f'(u)(u-1) - f(u)}{(u-1)^2}$; $f(u) = u \cdot \exp\left\{\frac{\beta\gamma(1-u)}{1+\beta(1-u)}\right\}$ } Wolfram Alpha

$$\Rightarrow \frac{\exp\left\{\frac{\beta\gamma(1-u)}{1+\beta(1-u)}\right\}\left[-(\gamma+2)\beta u + \beta^2(u-1)^2 + 2\beta + 1\right]}{(1-\beta(u-1))^2}(u-1) - u\cdot\exp\left\{\frac{\gamma\beta(1-u)}{1+\beta(1-u)}\right\} < 0$$

$$\Rightarrow \frac{\left[-(\gamma+2)\beta u + \beta^2(u-1)^2 + 2\beta + 1\right](u-1)}{(1-\beta(u-1))^2} - u < 0$$

$$\Rightarrow \left[-(\gamma+2)\beta u + \beta^2(u-1)^2 + 2\beta + 1\right](u-1) - u\cdot(1-\beta(u-1))^2 < 0$$

$$\left[-(\gamma+2)\beta u + \beta^2(u^2+1-2u) + 2\beta + 1\right](u-1) - u\left(1 + \beta^2(u-1)^2 - 2\beta(u-1)\right) < 0$$

$$\Rightarrow \left[-\beta\gamma u - 2\beta u + u^2\beta^2 + \beta^2 - 2u\beta^2 + 2\beta + 1\right](u-1) - \left(u + u\beta^2(u-1)^2 - 2\beta u(u-1)\right) < 0$$

$$\left[-\beta\gamma u^2 - 2\beta u^2 + u^3\beta^2 + u\beta^2 - 2u^2\beta^2 + 2u\beta + u + \beta\gamma u + 2\beta u - u^2\beta^2 - \beta^2 + 2u\beta^2 - 2\beta - 1 - u - u^3\beta^2 - u\beta^2 + 2u^2\beta^2 - 2\beta u^2 + 2\beta u\right] < 0$$

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This can be solved to get condition for inequality relating β, γ, u using the roots of cubic to less than 0